Name: ……

END OF TERM 1 EXAM

P525/1

CHEMISTRY

PAPER 1

APRIL/MAY 2018

2 ¾ HOURS

Uganda advanced certificate of education

Chemistry

Paper 1

2hours 45 minutes

Instructions to candidates:

- ✓ *Answer all questions in section A and six questions in section B.*
- ✓ *The questions must be answered in the spaces provided.*
- ✓ *The periodic table is provided at the back of the paper.*
- ✓ *Illustrate your answers with equations where applicable.*

For examiner's use only																
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17

SECTION A

1. The pH of a 0.2M solution of a weak acid, HA is 4.3
   a) What is meant by the term
      i) Weak acid (1mark)

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      ii) pH of a solution? (1mark)

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   b) Write the:
      i) Equation for dissociation of the weak acid, HA in water . (1mark)

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      ii) Expression for the acid dissociation constant Ka of the acid, HA. (1 mark)

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   c) Calculate the
      i) Hydrogen ion concentration (1mark)

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      ii) acid dissociation constant, K_a of HA. (1marks)

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2. a) State graham's law of diffusion.
   1mark
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(b) A hydrocarbon C_nH_{2n} diffuses at $250cm^3$ in 10 seconds and oxygen gas diffuses at $716cm^3$ in 25 seconds under the same conditions. Calculate the molecular mass of CnH2n and deduce its molecular formula
3marks

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3. a) Complete the following equations for nuclear reactions

(i) $^{239}_{94}Pu + ^{1}_{1}n \longrightarrow ^{91}_{39}Y + ^{146}_{55}Cs +$…… *(01 mark)*

(ii) $^{239}_{92}U \longrightarrow ^{239}_{98}NP +$…… *(01 mark)*

(iii) $^{226}_{88}Ra \longrightarrow ^{4}_{2}He +$…….. *(01 mark)*

iii) When a radioactive isotope was stored for 42 days, it retained $^{1}/_{8}$th of its original activity.
Calculate the half – life of the isotope. (02 marks)

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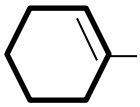
4. Complete the following equations and in each case name the main organic product. (4 ½ marks)

$$CH_3CH{=}CH_2 \xrightarrow[\text{warm}]{H^+/H_2O}$$

a)

Name: ……

b)  Br $\xrightarrow{\text{HBr}}$

Name……

c) CH3 $\xrightarrow{CrO_2Cl_2}$

Name:……

5. State what is observed and write the equation for the reaction which takes place when;
   a) Ethyne is bubbled through ammoniacal silver nitrate solution
      (2 marks)

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   b) Sodium carbonate is added to an aqueous solution of aluminium sulphate (2 marks)

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   c) Acidified potassium permanganate solution is added to an aqueous solution of sodium oxalate (2 marks).

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6. Sketch and give the name of the shape of each of the following species.
   3marks

Species	Sketch	Name
a) $CHCl_3$		

b) ClO_2^-		
c) CO_3^{2-}		
d) H_2O		
e) SF_4		

7. 50.0 g of substance Y was dissolved in water to make 1000 cm^3 of solution. If the partition coefficient of Y between water and ether is 0.2, calculate the mass of Y extracted by shaking the 1000 cm^3 of solution with:

(a) 500 cm^3 of ether. (*02 marks)*

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(b) two successive portions of 250 cm^3 of ether. (*02 marks)*

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d) Comment on the mass of Y extracted in a) and b).

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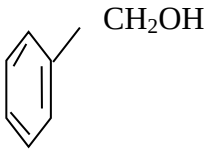
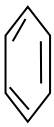
8. Show how the following conversions could be carried out.

(a) 

to  (2 ½ marks)

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(b) $CH_3CH_2CH_2OH$ to $CH_3CH(OH)CH_3$ (2 ½ marks)

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9. (a) (i Write the electronic configuration of aluminium *(1 ½ marks)*

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(ii) State the type of bonding that exists in aluminium fluoride. *(1 mark)*

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(b) By using equations only, explain why aqueous solution of aluminium chloride is acidic. *(1 ½ marks)*

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SECTION B

10. A compound P contains 52.2% carbon, 13.0% hydrogen, the rest being oxygen.

(a) Determine the empirical formula of P.

(02 marks)

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(b) When vaporized, 0.1g of P occupied $78.8cm^3$ at 157^oC and a pressure of 740mmHg.

(i) Calculate the formula mass of P. *(2 ½ marks)*

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(ii) Determine the molecular formula of P. *(1 ½ marks)*

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(iii) Write the structural formula of all the possible isomers of P.*(01 mark)*

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(c) P does not react with sodium metal. Identify P. *(0 ½ mark)*

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(d) Write an equation to show how P can be prepared from methanol. *(1 ½ marks)*

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11. a) 0.20 moles of phosphorous (v) chloride decomposed at 600k in a vessel of 5.00dm3. the equilibrium mixture is found to contain 0.08 moles of chlorine

$PCl_{5(g)} \rightleftharpoons PCl_{3(g)} + Cl_{2(g)}$; $\Delta H = -96kj/mol$

i) Write the expression for K_p and K_c of the reaction stating the units 2marks

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ii) Calculate the value of K_c (3 marks)

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iii) State and explain the effect on the value of K_c and position of equilibrium when the temperature of the vessel is raised to 800k. 3 marks

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iv) State what is observed and write the equation for the reaction taking place when phosphorous (v) chloride is added to ethanol. (1 ½ marks)

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12. (a) Ethylethanoate can be hydrolyzed in presence of a base as shown below.

$$CH_3COOCH_2CH_3 \xrightarrow[\text{Heat}]{H_2O/OH_{(aq)}} A + B$$

Identify substances A and B 1 mark

A:

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B:

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(b) The table below shows results that were obtained during hydrolysis of ethyl ethanoate

Experiment	$[CH_3COOCH_2CH_3]$ ($moldm^{-3}$)	$[\bar{O}H]$ ($moldm^{-3}$)	Initial rate ($moldm^{-3}s^{-1}$)
1	0.096	0.096	$1.13x10^{-3}$
2	0.048	0.192	$1.13x10^{-3}$
3	0.024	0.192	$5.65x10^{-4}$

i) What is meant by the term "order of reaction"
1mark

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ii) Deduce order of reaction with respect to 1mark@

$[\bar{O}H]$:

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[$CH_3COOCH_2CH_3$]:

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iii) Write rate equation and the overall order of reaction 1 marks

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iv) Calculate the rate constant 1 marks

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(c) Explain how a catalyst works when altering the activation energy of a reaction 2marks

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13. (a) state Hess's law of heat summation
1mark

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(b) Define the term standard enthalpy of formation
3marks

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(c) The following standard enthalpies of formation data refer to second stage in the extraction of titanium

	$TiCl_{4(g)}$	$Na_{(l)}$	$NaCl_{(s)}$	$Ti_{(s)}$
ΔH_F^{θ}/KJmol^{-1}	-720	+3	-411	0

i) State why the value for the standard enthalpy of formation of $Na_{(l)}$ is not zero 1mark

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ii) Use the data from the table to calculate the value for the standard enthalpy change of the following

$TiCl_{4\,(g)} + Na_{(l)} \longrightarrow NaCl_{(s)} + Ti_{(s)}$

3marks

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iii) State the role of sodium in this reaction 1mark

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14. Write equations to show how the following compounds can be synthesized

(a) CH_3CHO (from 1,2 – dibromoethane) (3 marks)

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(b) CH_3CO_2H(from iodomethane) (2 ½ marks)

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(c) $CH_3CO_2CH_3$ (from ethene) (3 ½ marks)

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15. Write the name and formula of an ore of aluminium. *(1 mark)*

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(b) During the extraction of aluminium, the ore is first treated with sodium hydroxide, followed by aluminium hydroxide.

(i) State the purpose of adding sodium hydroxide.

(1 mark)

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(ii) Write equation for the reaction between the ore and sodium hydroxide. *(1 ½ marks)*

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(iii) What is the purpose of adding aluminium hydroxide? *(1 mark)*

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(c) Briefly explain how aluminium can be obtained after the ore has been treated as in (b) *(3 marks)*

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(d) Carbon dioxide was used instead of aluminium hydroxide in (b). Write equation for the reaction that took place. *(1 ½ marks)*

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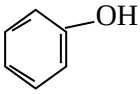
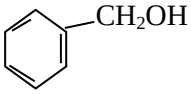
16. Name a reagent that can be used to distinguish between each of the following pairs of compounds. State what would be observed if the compounds are treated with the reagent:

(a) 

and 

(3 marks)

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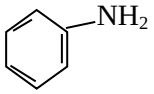
(b)  and $CH_3CH_2CH_2\ CH_2\ CH_2\ CH_2NH_2$ *(3 marks)*

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(c) $CH_3CH_2CH{=}CH_2$ and $CH_3CH_2C{\equiv}CH$ *(3 marks)*

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17. Nitric acid and water are miscible in all proportions. They form a maximum boiling point mixture containing 68% nitric acid which boils at 108°C.

(a) Sketch a labelled boiling point – composition diagram for mixture of nitric acid and water. *(3 marks)*

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(b) Explain the factors that lead to nitric acid and water forming a maximum boiling point mixture.

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(c) Describe briefly what happens when a mixture containing 20% nitric acid is distilled. *(2 marks)*

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(d) Give one method by which pure methanoic acid can be obtained from the maximum boiling point mixture. *(1 mark)*

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END